

CONTACT INFORMATION

✉ E-mail: bm452@sussex.ac.uk

EDUCATION

University of Sussex 2023-

- PhD in Informatics
Thesis: Considering spatial embeddings and synaptic delays for neural networks in neuromorphic computing

ELTE Eötvös Loránd University 2020-2023

- M.Sc. in Artificial Intelligence
Thesis: Extending Sparse Dictionary Learning Methods for Adversarial Robustness
Final Grade: *Excellent*

Humboldt University Berlin 2022

- Exchange semester with ERASMUS

ELTE Eötvös Loránd University 2016-2020

- B.Sc. in Computer Science, specialization in modelling
Thesis: Generating Symbolic Music using Artificial Neural Networks
Final Grade: *Good*

Johannes Kepler University Linz 2019

- Exchange semester with CEEPUS

EXPERIENCE

Research

- **Sussex AI**
◦ Delay learning and structural plasticity 2023-
- **MARCS Institute**
◦ Visiting researcher at the ICNS 2026
- **Cold Spring Harbor Laboratory**
◦ NeuroAI research intern 2026
- **Centrum Wiskunde & Informatica**
◦ Visiting researcher with Sander Bohte 2025
- **The Alan Turing Institute**
◦ NeuroAI and backdoor reinforcement learning 2024-2025
- **Neural Information Processing Group**
◦ Sparse approximation for adversarial robustness 2021-2022
◦ Multimodal emotion estimation using transformer architectures 2021

Teaching

- **Neuromatch Academy**
◦ NeuroAI 2025
- **University of Sussex**
◦ Bio-inspired AI 2024
- **ELTE Eötvös Loránd University**
◦ Algorithms and Data Structures 2020
◦ Object-oriented programming in C++ 2018

Software development

- **evosoft**
◦ C++ Software Engineer 2019-2020

SKILLS	• <i>Programming Languages</i>: C/C++, Python, Prolog • <i>Softwares/Libraries</i>: NumPy, Keras, PyTorch, OpenCV, Microsoft Office, Latex • <i>Language Skills</i>: English: C1 (Advanced), German: A2 (Elementary)	
AWARDS AND FELLOWSHIPS	• MARCS International Visiting Scholar Fellowship • The Alan Turing Institute - PhD enrichment scheme • be.AI Leverhulme Doctoral Scholarship • Nokia Young Scientist Award • TDK - 1st prize	2026 2024 2023 2023 2022
CERTIFICATES	• Associate Fellow in Advanced HE • Neuromatch Computational Neuroscience Summer School	2024 2023
PUBLICATIONS	• Journal ◦ Mészáros, B., Knight, J., & Nowotny, T. <i>Nature Communications</i>. Efficient Event-based Delay Learning in Spiking Neural Networks ◦ Ghosh, M., Habashy, K. G., De Santis, F., Fiers, T., Ercelik, D. F., Mészáros, B., ... & Goodman, D. F. <i>eNeuro</i>. Spiking neural network models of interaural time difference extraction via a massively collaborative process ◦ Mészáros, B., Knight, J., & Nowotny, T. <i>Frontiers in Computational Neuroscience</i>. Learning Delays Through Gradients and Structure: Emergence of Spatiotemporal Patterns in Spiking Neural Networks. • Conference ◦ Mészáros, B., Knight, J. C., Timcheck, J., & Nowotny, T. <i>International Conference on Neuromorphic Systems (ICONS)</i>. A Complete Pipeline for deploying SNNs with Synaptic Delays on Loihi 2 ◦ Shoesmith, T., Knight, J. C., Mészáros, B., Timcheck, J., & Nowotny, T. <i>Neuro Inspired Computational Elements (NICE)</i>. Eventprop training for efficient neuromorphic applications ◦ Szeghy, D., Aslan, M., Fóthi, Á., Mészáros, B., Milacski, Z. Á., & Lőrincz, A. <i>International Conference on Deep Learning Theory and Applications (DeLTA)</i>. Structural Extensions of Basis Pursuit: Guarantees on Adversarial Robustness. • Submitted ◦ Mészáros, B., Knight, J. C., Akarca, D., & Nowotny, T. Space as Time Through Neuron Position Learning. Submitted to <i>Nature Machine Intelligence</i>. • In preparation ◦ Pehle, C., Mészáros, B., & Zador, A. In preparation.	
ACADEMIC OUTREACH	• Talks ◦ Budapest Computational Neuroscience Forum ◦ Spiking Neural Networks as Universal Function Approximators ◦ International Workshop on Reliable and Sustainable Neuromorphic Hardware ◦ Neuro Inspired Computational Elements Conference ◦ Sensory Neuroscience and Computation centre seminar ◦ Robots in Disguise: Fundamental AI Reading Group ◦ MPIE Conference ◦ OTDK - National Scientific Student Body Conference • Posters ◦ Spiking Neural Networks as Universal Function Approximators ◦ 33rd annual Computational Neuroscience Meeting ◦ The Fifth International Convention on the Mathematics Of Neuroscience and AI ◦ International Conference on Neuromorphic Computing and Engineering	2026 2025 2025 2025 2025 2024 2024 2023 2024 2024 2024 2024

ADDITIONAL	• Welfare at FOLD	2025-
	• Volunteer work	
	◦ A Város Mindenkié	
	◦ Heti Betevő	
	◦ Organizing AI themed artisan workshops for layman	
	• Sport activity	
	◦ Sabre fencing - 4 times national champion, European Championship 4th place	2005-2016
	• Music production, Songwriting, DJing	
	◦ Debut EP forthcoming on SCORDISCI	2026